

OU fits to the gain-modulated RNNs: RT-and-choice fits stay leaky at every threshold; the same accumulator fitted to choice given the evidence recovers the transition

Ivan Grahek, 24 September 2026. Round 3 summary; full write-up docs/OU_fits_to_gain_RNNs_round2.pdf, code and RESULTS.md on branch round2-report of rnn_hssm. Prediction: gain 0.8 / 1.0 / 1.2 = leaky / near-perfect / unstable integration, so the OU leak should go positive / zero / negative. Convention throughout: ssm-simulators' update is $x += (v - g \cdot x) \cdot dt + \text{noise}$, so **positive g is leaky**, $g = -\lambda$ in Brunton's notation. All intervals are 95 % unless stated.

1. Control: HSSM OU fitted to RT and choice, at four thresholds on dv

The networks read out their choice at 750 ms and do not produce RTs. Every RT below is a threshold we place on the decision variable dv; trials that never reach it are dropped (HSSM has no censored likelihood). At a low threshold the crossing is driven by fast fluctuations of the readout; near the committed attractors ($|dv| \approx 3$) the slow choice mode governs it. Fit: pretrained OU likelihood, 20 networks pooled per gain, RTs $\times 10$, non-decision time fixed at offset + minimum RT, no lapse, $v \sim 1 +$ signed coherence, a , z and g shared across coherences, uniform priors over the likelihood's box. All 12 fits converged (R-hat 1.00, one divergence in 48 000 draws).

At threshold 1.5 the posterior for g sits on the likelihood's +1 edge (stretched units) at gains 0.8 and 1.0 for every stretch from $k = 8$ to 24, so its native value equals k and only the sign is identified; 120 per-network fits at that threshold give $g > 0$ in 120 of 120. The pipeline can see instability: with the theoretical g planted in simulated data it returns $g < 0$ at gains 1.0 and 1.2 (sign recovered in 6 of 6).

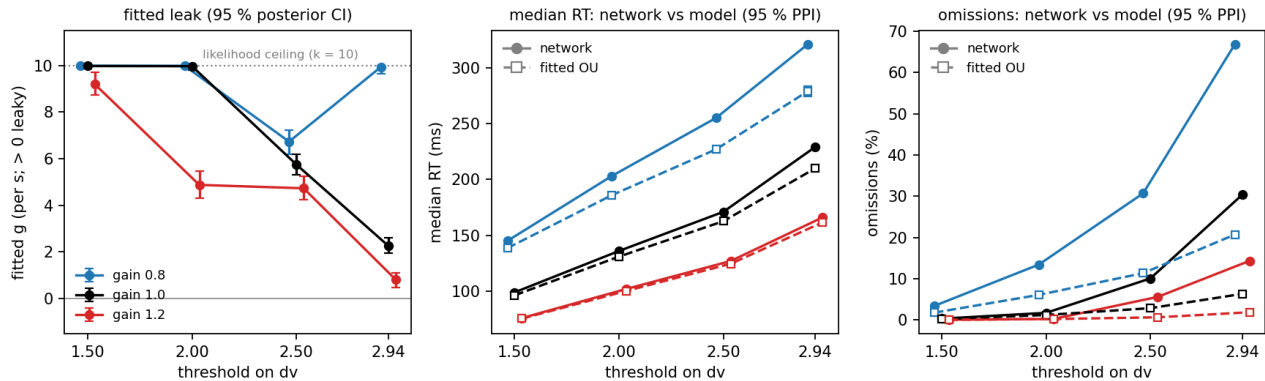


Figure 1. How the fit changes with the threshold. Left: fitted g (posterior median, 95 % interval); it falls and orders by gain as the theory predicts but never becomes negative, and gain 0.8 returns to the ceiling at 2.94. Middle and right: median RT of crossing trials and omission rate, network (filled) against the model's 95 % posterior-predictive interval (open; 40 posterior draws, one replicate dataset of the data's size per draw). The predictive intervals are narrower than the markers (40 000 trials per fit). At 1.5 the model is within a few ms and points (e.g. gain 0.8: omissions 3.4 % vs 1.6–1.9 %); at 2.94 it is too fast and omits far too few trials for the leaky networks (gain 0.8: median RT 321 vs 275–283 ms, omissions 67 % vs 20–21 %).

The decisive check drives each fitted model with the networks' own evidence streams and computes the psychophysical kernel (Figure 4). A higher threshold moves the model toward the network and brings out the ordering across gain: at 2.94 the model's slopes are +0.006, -0.024 and -0.043 against the networks' +0.039, +0.002 and -0.026 . But the fitted bound is still reached before the deadline on 77 to 98 % of trials, so the model stays shifted toward primacy and has the wrong sign at gains 0.8 and 1.0. Its bin-by-bin kernel error falls from 0.151 at threshold 1.5 to 0.065 at 2.94, still six times that of the choice-given-evidence fit in Section 2 (0.011) — and at 2.94 the RT fit has already failed for gains 0.8 and 1.0.

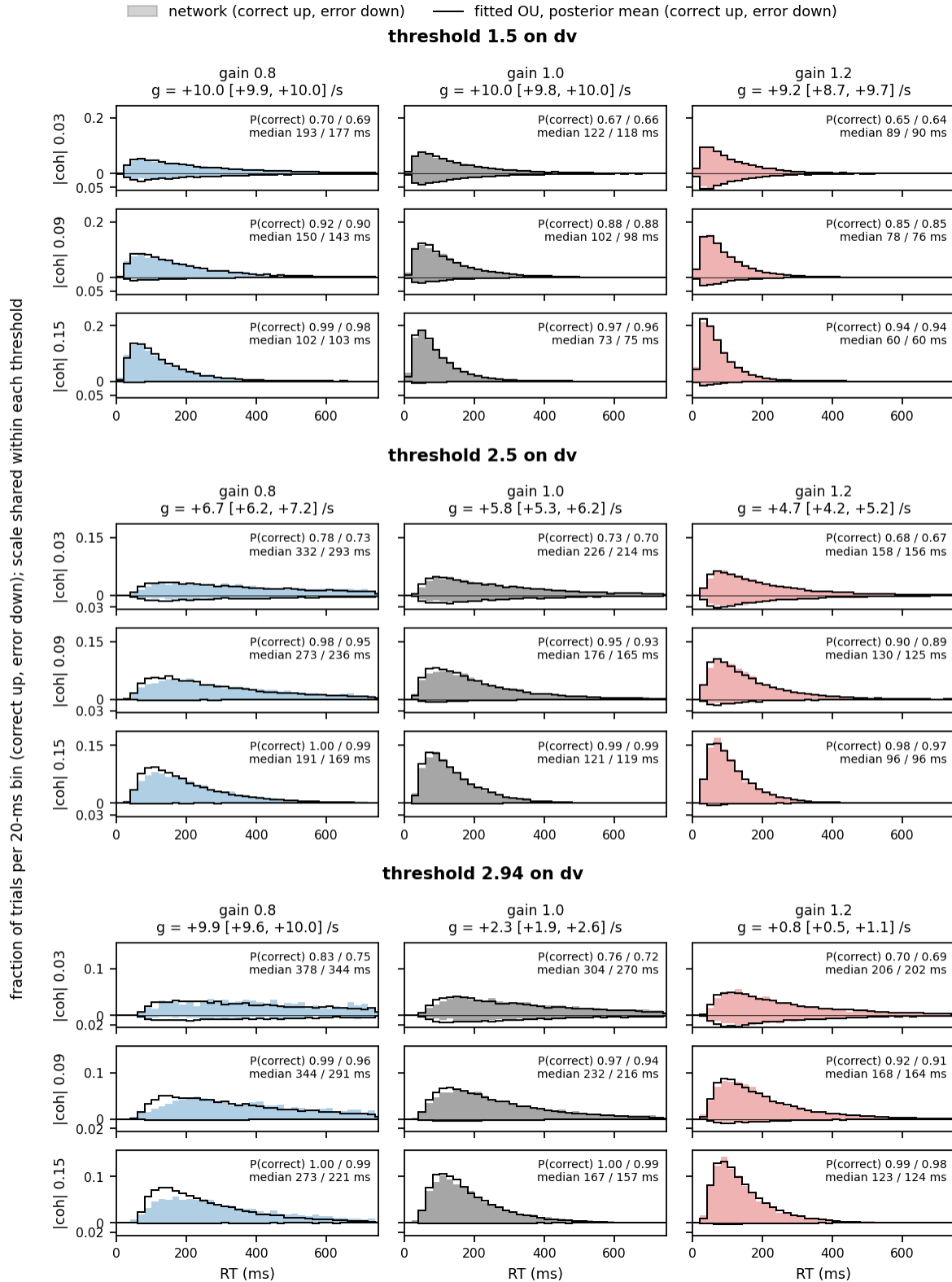


Figure 2. RT posterior predictive by threshold (blocks), [coherence] (rows) and gain (columns). Fraction of trials per 20-ms bin, correct up and errors down (the two sum to one), on a scale shared within each threshold block; network (filled) vs fitted OU at its posterior mean (line). Column titles: native g with 95 % posterior interval; insets: accuracy and median RT, network / model. At 1.5 the fit is close everywhere; at 2.5 and 2.94 it misses the leaky network's slow, flat RT distribution at every coherence, while gain 1.2 stays well described.

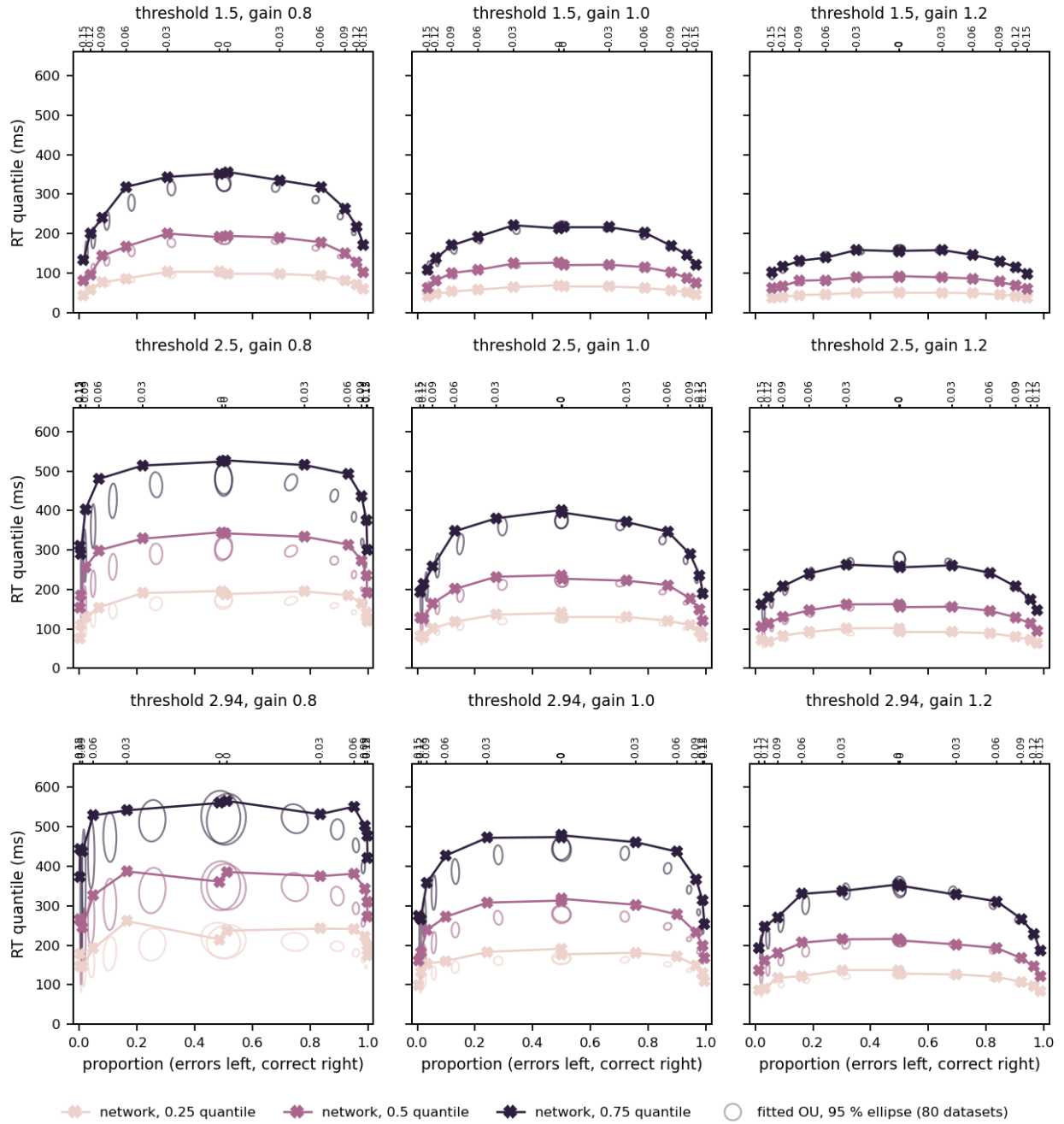


Figure 3. Quantile-probability plot, drawn like HSSM's `plot_quantile_probability` with ellipses (the call in `model/fit_ornstein.py`). Rows: threshold; columns: gain; each panel holds all six [coherence] levels (top axis). x: proportion of that condition's trials that are correct (right) or errors (left); y: RT quantiles 0.25, 0.5 and 0.75 (HSSM default). Network: X markers joined per quantile. Fitted OU: 95 % ellipse over 80 posterior-predictive datasets (HSSM default of 20 draws per chain $\times$ 4 chains), each with one simulated trial per data trial; the one departure from HSSM is that simulated trials missing the 750 ms horizon are dropped, as in the data. At 1.5 the ellipses sit on the network's quantiles for gains 1.0 and 1.2 and within about 30 ms for gain 0.8. At 2.5 and 2.94 the model's correct-trial quantiles for gain 0.8 fall 11–85 ms below the network's at every coherence and its accuracy is 3–4 points lower; gain 1.0 is off by up to 42 ms at 2.94; gain 1.2 stays within about 8 ms at every threshold.

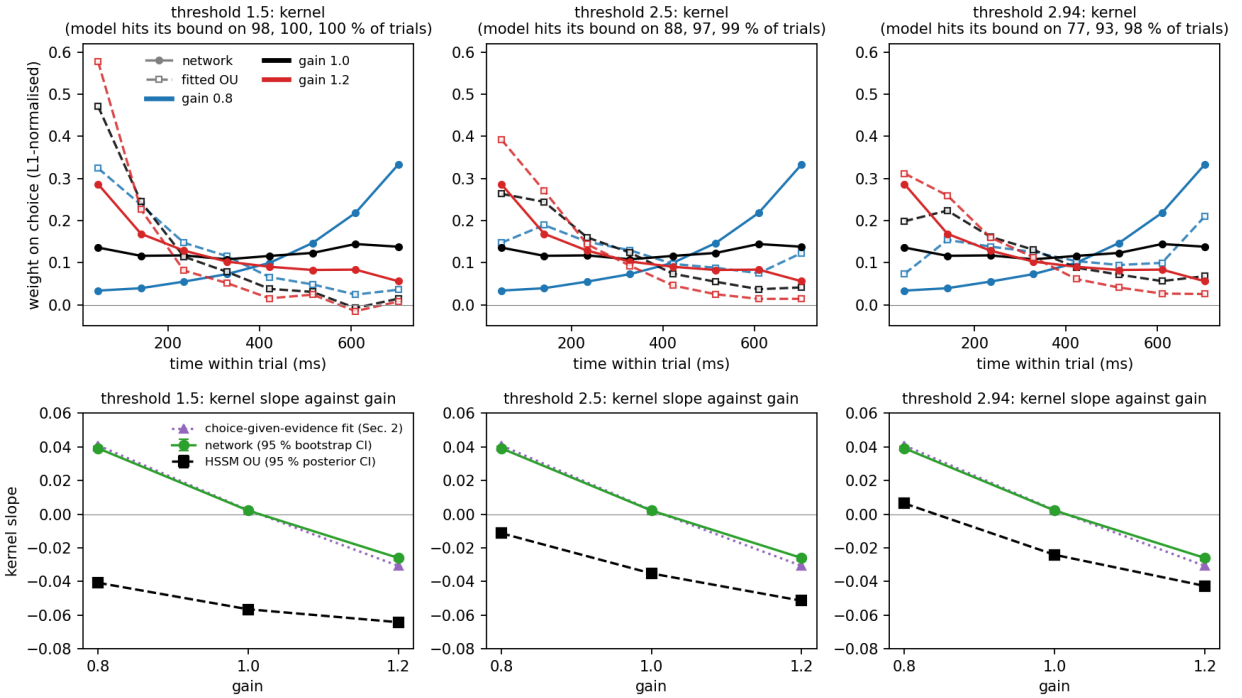


Figure 4. Kernel check at thresholds 1.5, 2.5 and 2.94. Top: kernels, network (solid) vs the fitted OU with its fitted bound driven by the same evidence (dashed). Bottom: kernel slope against gain; network with 95 % bootstrap interval over trials, HSSM OU with 95 % posterior interval over 40 draws, and the choice-given-evidence fit of Section 2 for reference. Both intervals are about ± 0.001 , smaller than the markers.

2. Result: the same OU fitted to each trial's choice given its evidence stream (Brunton-style, no RTs)

$a_{t+1} = a_t + (v e_t - g a_t) dt + \sqrt{dt} \xi_t$, $dt = 1$ ms, choice = sign(a_{750}); e_t is the evidence the network received. Without a bound a_t is Gaussian given the evidence, so each choice has an exact likelihood; fitted by maximum likelihood and NUTS per network and gain (60 fits) and pooled.

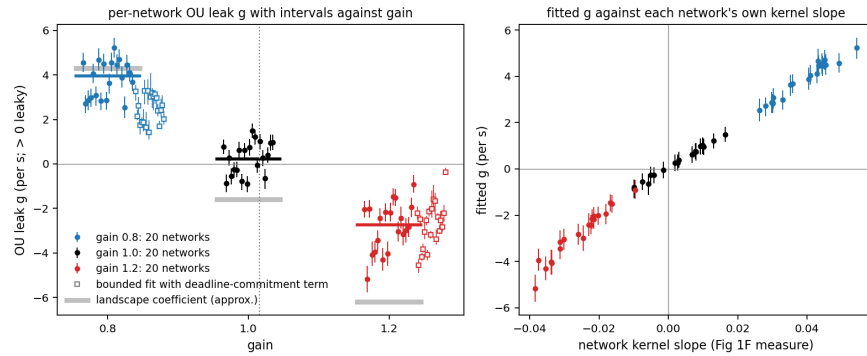


Figure 5. Left: fitted g per network with 94 % HDIs (filled), pooled fit (bars), bounded fit with the deadline-commitment term (open squares), landscape drift coefficient (gray), kernel zero crossing of Fig 1F at gain 1.016 (dotted). Right: g against each network's own kernel slope, $r = 0.996$.

gain	median g across networks (per s) [IQR]	intervals excluding 0 on the predicted side	pooled g [94 % HDI]	landscape coefficient (approx.)
0.8	+4.13 [+3.02, +4.63]	20 of 20	+3.95 [+3.83, +4.06]	+4.3
1.0	+0.32 [-0.34, +0.79]	10 positive, 5 negative	+0.23 [+0.15, +0.32]	-1.6
1.2	-2.63 [-3.56, -2.03]	20 of 20	-2.73 [-2.83, -2.62]	-6.2

Table 1. Leaky $\rightarrow$ near zero $\rightarrow$ unstable, in every network at the extreme gains. Zero crossing of g against gain: 1.016 (Fig 1F kernel: 1.016). Recovery on real evidence with fitted or theoretical g planted: 60 of 60 signs, 56 of 60 intervals cover the truth.

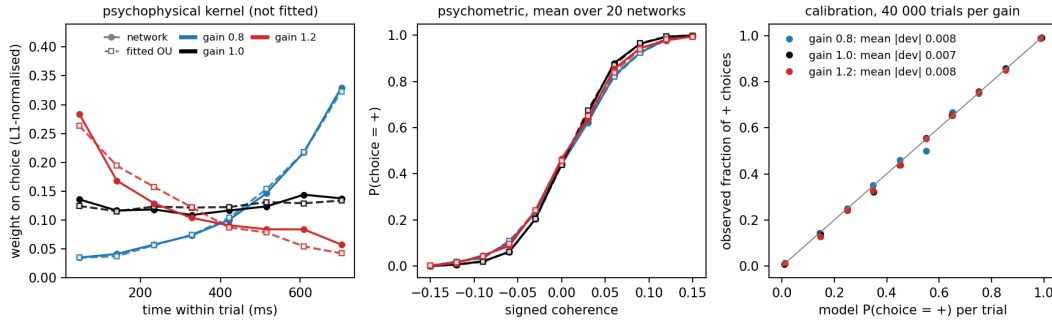


Figure 6. Posterior predictive. Left: the fit reproduces the psychophysical kernel it was not fitted to (slopes $+0.041 / +0.002 / -0.031$ vs network $+0.039 / +0.002 / -0.026$). Middle: $P(\text{choice})$ by signed coherence, within 0.015 everywhere. Right: per-trial predicted probability against observed choice fraction.

3. What we learned

The RT here is a readout we impose on dv, not something the network does, and what the RT-and-choice fit measures depends on where we put it. At a low threshold it is crossed by the fast readout fluctuations: the OU fits those RTs almost perfectly by putting its one leak term on their decay, leaky and at the ceiling at every gain, with the wrong kernel. Raising the threshold lets the slow dynamics show through: g falls and orders by gain and the kernel ordering appears. But g never becomes negative, and at the thresholds where the ordering appears the one-timescale linear OU with a constant bound no longer fits the leaky network's RTs or omissions.

Conditioning the same accumulator on each trial's evidence, and fitting only the choice the network actually makes, gives the predicted sign pattern per network with intervals and reproduces the kernel an order of magnitude better. A joint RT-and-choice model of these networks would need at least a second, fast timescale and a likelihood that handles non-crossers; the Brody lab's Fokker-Planck grid likelihood is the natural route for both. Caveats for the result: the magnitude at gain 0.8 depends on how a sticky bound is constrained ($+4.1$ free, $+2.5$ with the deadline commitment fraction; the sign does not change), and the linear OU under-reports the repulsion at gain 1.2 (-2.6 vs -6.2) because it cannot saturate into wells.